

RESUME SCREENING & APPLICATION TRACKING SYSTEM

Introduction

The Resume Screening & Application Tracking System is a web-based application developed using Spring Boot and H2 Database. It helps recruiters automate the recruitment process by analyzing resumes and managing candidate applications efficiently. The system allows candidates to upload PDF resumes, extracts resume content using Apache PDFBox, calculates skill-matching scores, assigns application status automatically, and stores all information in a centralized database. Recruiters can view, search, update, download, and manage candidate applications through a user-friendly dashboard.

Objective

- To automate the resume screening process.
- To extract and analyze resume content from PDF files.
- To compare candidate skills with predefined job requirements.
- To calculate candidate matching scores automatically.
- To assign application status based on screening results.
- To provide an efficient application tracking mechanism.

Technologies Used

- Java 17
- Spring Boot
- Spring Data JPA
- H2 Database
- Apache PDFBox
- HTML5
- CSS3
- JavaScript
- Bootstrap 5
- Maven

Features

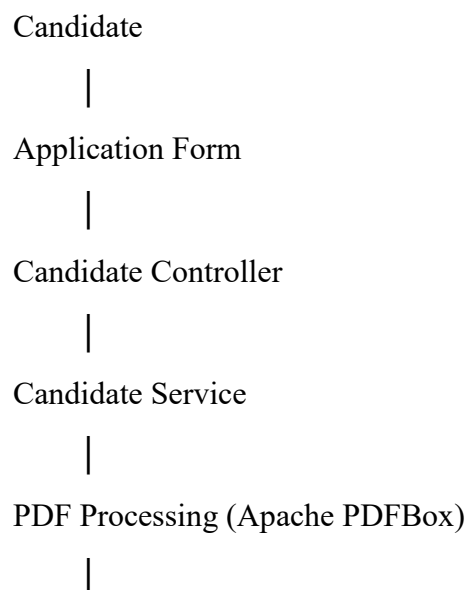
- Candidate Registration
- Resume PDF Upload

- PDF Text Extraction
- Automated Skill Matching
- Resume Scoring System
- Automatic Status Assignment
- Recruiter Dashboard
- Candidate Search Functionality
- Resume Download Option
- Status Management
- Dashboard Statistics

System Flow

1. Candidate submits application details.
2. Resume PDF is uploaded.
3. Resume is stored in the uploads folder.
4. PDFBox extracts resume text.
5. Extracted text is compared with required skills.
6. Matching score is calculated.
7. Status is assigned automatically.
8. Candidate information is stored in H2 Database.
9. Recruiter manages applications through the dashboard.

Architecture



Resume Screening Engine

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Candidate Repository

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H2 Database

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Recruiter Dashboard

Future Enhancements

- AI-Based Resume Ranking
- Email Notification System
- Interview Scheduling Module
- Multiple Job Position Support
- Cloud Storage Integration
- Advanced Recruitment Analytics

Conclusion

The Resume Screening & Application Tracking System successfully automates resume screening and application management. The system reduces manual effort, improves recruitment efficiency, and provides a structured approach for evaluating candidates. The project demonstrates the practical implementation of file handling, PDF processing, database operations, and CRUD functionalities using Spring Boot.